Sedimentree

Introduction

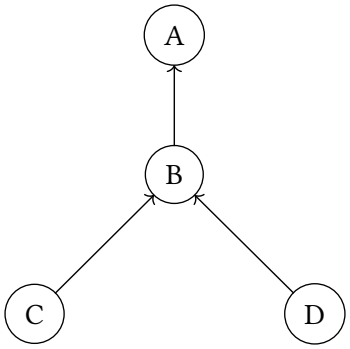
`stockfoil` is a library for building CRDTs which are designed to be replicated as a "sedimentree". Thus to understand a lot of what `stockfoil` is doing we need to understand what a sedimentree is and what it is trying to achieve.

Sedimentrees

A sedimentree is a data structure designed to allow two peers to synchronize a commit graph without either peer having to load the entire commit graph and do complex ancestry computations, whilst still enable incremental sync for mostly caught-up peers.

Syncing Commit Graphs

Here's a commit graph (arrows point from childrent to parents).



There are various approaches to synchronizing this commit graph. The strategy sedimentree takes is to

Non-boundary Fragment Dependencies

The commits which are inside a fragment are defined as those which are ancestors of the head commit, but not ancestors of the boundary. On the other hand, the boundary itself is defined by walking backwards up the graph from the head commit until every walk has terminated in a commit at the same depth as the head.

This is actually a bit asymmetric and it leads to this strange scenario where you have a commit graph which looks like this:

```
  A
 / \
B   C
 \ /
  D
```

And `D` and `B` and `A` are all at the same depth. Then the head of the fragment is `D` and the boundary is `A`. What's weird is that the commit `A` is not actually a part of the